IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

Friday, 17 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n : Weak Inversion Slope factor.

g_m*: Transconductance.*

1.

a) Describe in your own words what the benefits are of using current mode rather than voltage mode.

[5]

b) *"Analogue circuits are better for signal processing than digital circuits"*. Prove this statement from first principle MOSFET noise equations and discuss in your own words for which scenarios this is true.

[10]

c) For a given mixed mode sensing system, analogue signals need to be digitized for data transmission. What is the best strategy to implement such a system which minimizes power consumption? You are required to describe this in your own words and show an example of an optimum system.

[5]

2. This question concerns translinear circuits.

a) State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion and discuss the limitations. Give all assumptions you make.

[5]

b) For the circuit shown Figure 2.1, I_{X1} and I_{X2} are input currents whereas I_Z is the output current. Express the current I_Z as a function of the currents I_{X1} and I_{X2} and the transistor area K. When $I_{X1}=I_{X2}=I$, determine the value of the transistor area K for which $I_Z=I$.

[10]

c) Figure 2.2 shows a translinear analogue multiplier. Derive its transfer function and show how this circuit can be modified to improve its gain. Derive the differential gain for a two stage multiplier consisting of the improved circuit.

[5]

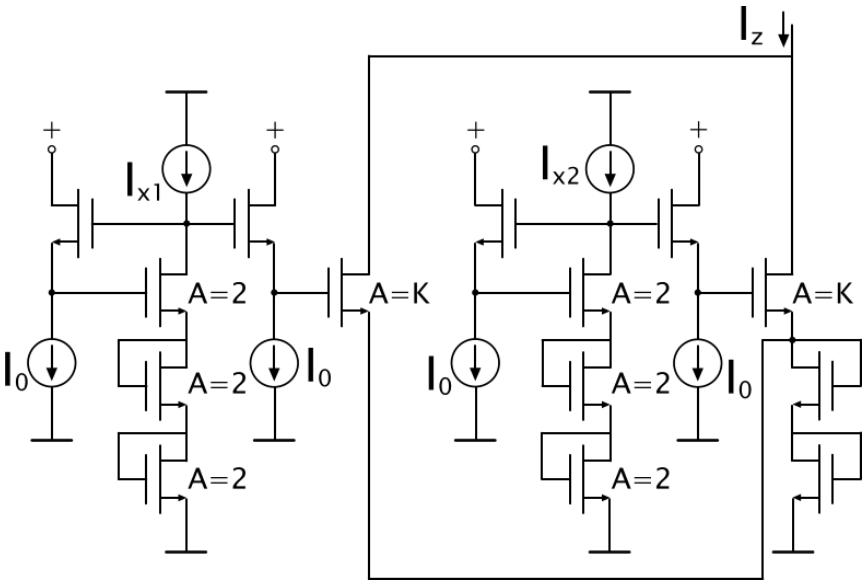


Figure 2.1

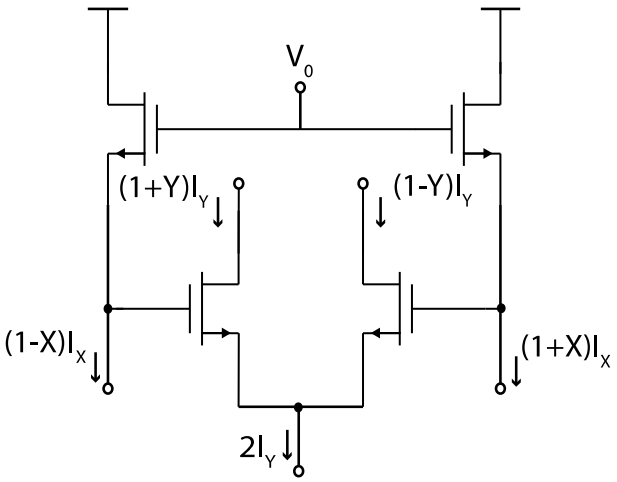


Figure 2.2

3.

a) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_T \tanh(V_c / nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f() composed of just I_{in}, I_1 and I_T.

[6]

b) The transfer function of a second order topology has been decomposed into the following state-space equations:

$$\dot{X_1} = -\left(\frac{\omega_0}{2Q}\right) X_1 - \omega_0 \left(1 - \frac{1}{4Q^2}\right) X_2 + \omega_0 U$$
$$\dot{X_2} = -\left(\frac{\omega_0}{2Q}\right) X_2 + \omega_0 X_1$$
$$Y_1 = X_1$$
$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs, X_1 and X_2 are the state-variables and U is the input.

i. Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

[4]

ii. By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[5]

iii. With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.

[5]

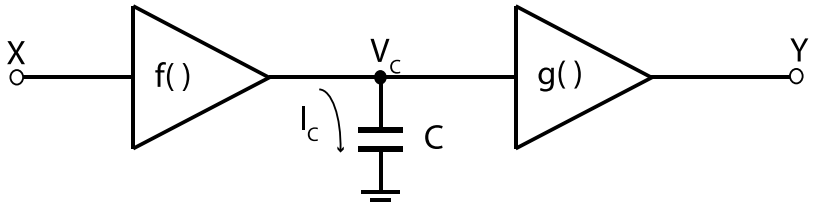


Figure 3.1

4.

a) Figure 4.1 depicts a switched current memory cell. Using this memory cell, design a unit delay cell proving a z^{-1} function and derive the equation for its operation.

[4]

b) Figure 4.2 depicts a FIR filter architecture. Using your unit delay cell from part 4.a sketch a full circuit schematic that implements this function. You may assume the current is unidirectional and use only pMOS current mirrors to copy the current. The coefficients $b_{0\text{-}3}$ must be tunable. Hint: they can be represented by current ratios I_{b0}/I_0, I_{b1}/I_0, I_{b2}/I_0 and I_{b3}/I_0, where I_0 is a MOSFET current in weak inversion.

[8]

c) Figure 4.3 shows a schematic of a switched current integrator and its associated timing diagram. Derive the transfer function of the switched current integrator and find its time constant. You may assume that the switches are ideal and that the transistors sizing is as follows: $\left(\frac{W}{L}\right)_{M_1} = \left(\frac{W}{L}\right)_{M_2}$, $\left(\frac{W}{L}\right)_{M_4} = b_4\left(\frac{W}{L}\right)_{M_1}$ and $\left(\frac{W}{L}\right)_{M_3} = b_3\left(\frac{W}{L}\right)_{M_1}$.

[8]

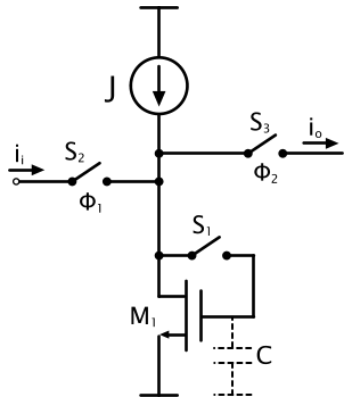


Figure 4.1

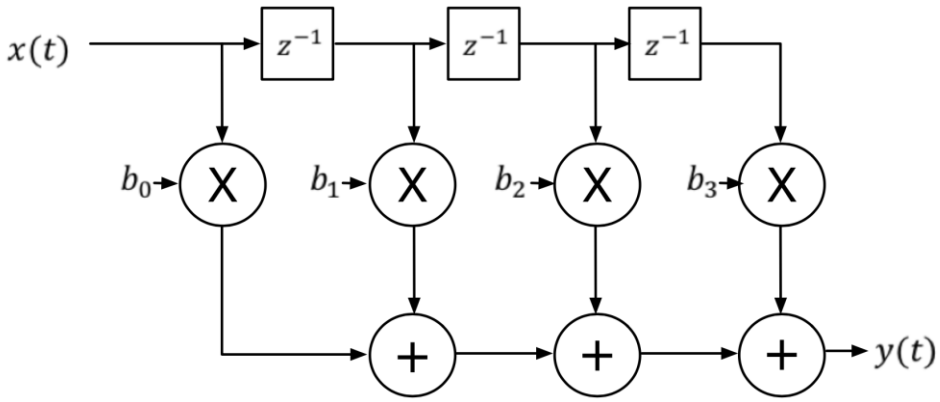


Figure 4.2

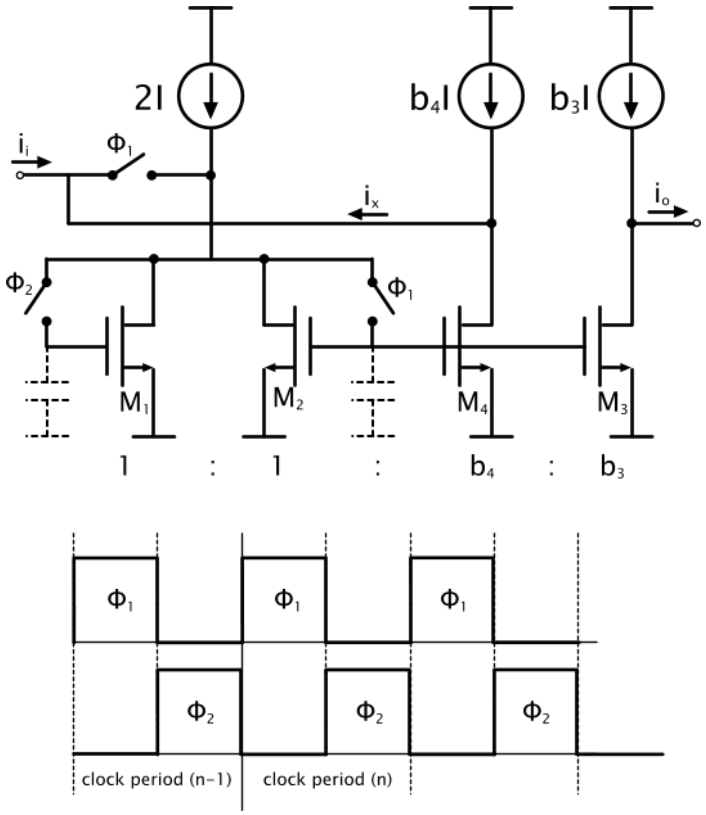

2I
b4I
b3I
ii
Φ1
ix
io
Φ2
Φ1
M1
M2
M4
M3
1 : 1 : b4 : b3
Φ1
Φ1
Φ1
Φ2
Φ2
Φ2
clock period (n-1)
clock period (n)

Figure 4.3

5. a) Figure 5.1 shows a NMOS sampling switch to be used in a switched capacitor circuit. The parameters of M1 are $(W/L)_1 = 30\mu m/1\mu m$, $C_{ox} = 17$ fF/μm^2, $V_{th} = 0.3$ V, CK = V_{DD} = 1.8 V and $C_H = 0.1$ pf. The total charge in the inversion layer, when the switch is on, is given by:

$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{th})$$

i) Explain the errors caused due to charge injection and clock feedthrough. Derive equations that show how the output voltage will change due to these errors, assuming the charge injection is split equally between the source and the drain.

[4]

ii) Using the output voltage equations derived in 5.a.i, calculate the total change in output voltage due to charge injection and the total change due to clock feed-through. Based on your result, specify which is more significant. You may assume the overlap capacitance $C_{ov} = 0.08$ fF/m and an input voltage $V_{in} = 1$ V.

[3]

iii) Show that the output voltage contribution due to charge injection results in a non-unity gain error and constant offset. Plot a graph indicating the value of the offset and the slope.

[3]

b) Figure 5.2 shows a circuit diagram of a differential switched capacitor non-inverting amplifier.

i) Explain the necessity of bottom plate sampling in this design and draw a schematic of the amplifier indicating which sides of the capacitors are the bottom plates.

[5]

ii) Derive the transfer function of the Figure 5.2 and also derive an equation to show the change in output voltage during phases Φ_1 and Φ_2.

[5]

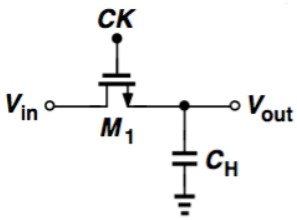


Figure 5.1

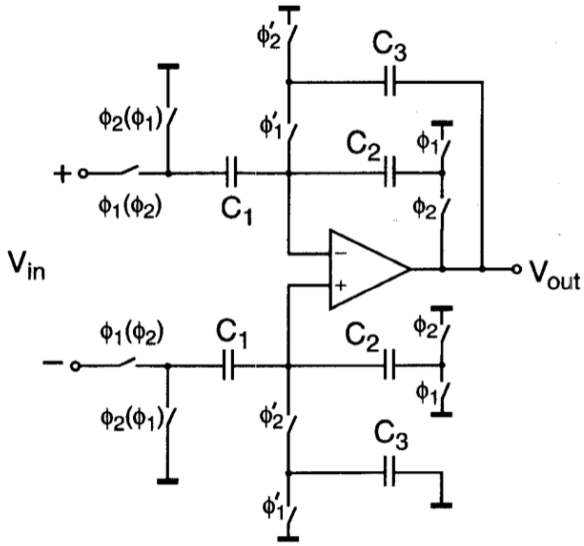


Figure 5.2

6. Figure 6.1 shows a measurement system that digitizes the product of two different signals. The architecture uses an operational amplifier, a current multiplying circuit and an analogue-to-digital converter to encode the detected voltage signal. The detected signals are first converted to currents through a voltage-to-current converter (V/I), then multiplied in current mode and finally converted to a spike train through an integrate-and-fire neuron (I/F). The 8 bit counter is used to count the number of spikes in a fixed period T_{rst}. The whole system works off a 1 V supply.

a) Describe a technique which allows detection of very small voltage signals which exist below the 1/f noise floor of the operational amplifier. Include diagrams showing the operation of this technique and how the frequency spectrum of the signal and noise is shaped in each stage.

[5]

b) Sketch a circuit for multiplying two currents. State the region of operation for the MOSFETs and derive the relationship between the output and the input. Identify a way to provide a gain to the multiplication.

[4]

c) Sketch an integrate-and-fire circuit which can be used in this design and describe its operation. Derive equations for the pulse width and frequency.

[5]

d) Given that the maximum output is limited to $I_{out} = 100$ nA, choose a suitable capacitor value for your integrate-and-fire circuit such that the spike frequency is always less than 5 kHz. You may assume the refractory current is $I_r = 1$ nA.

[4]

e) Calculate the maximum allowable reset time T_{rst} such that the counter never overflows for this current. How many bits are required when integrating over a long time constant $T_{rst} = 1\ s$?

[2]

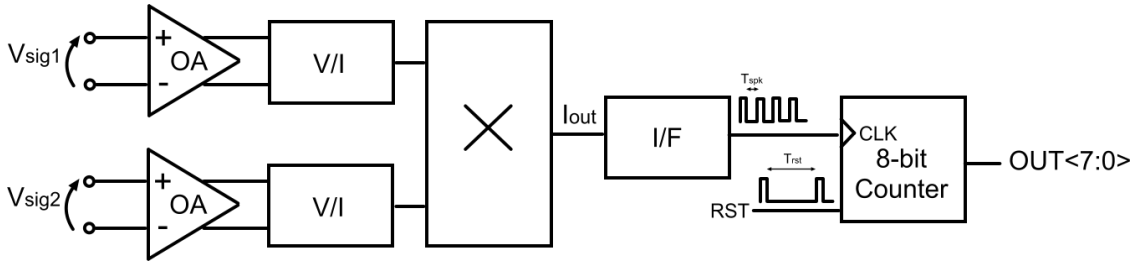


Figure 6.1